

Beyond Pass@1: Reliability–Cost Frontiers of Quantized Reasoning Models under Controlled Serving-Stack Shift

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Abstract

Post-training quantization is standard practice for reducing the prohibitive memory and latency footprints of Large Language Models (LLMs). While traditional compression literature evaluates short-form classification or QA benchmarks using single-seed pass@1 accuracy, reasoning LLMs generate long deliberation chains that introduce subtle, multi-step failure dynamics: token length inflation, termination anomalies, calibration drift, and shifted serving economics. In this work, we present a rigorous empirical study evaluating the joint reliability–calibration–cost frontier of quantized reasoning LLMs under a controlled, pinned serving stack (vLLM 0.7.0 eager execution on NVIDIA A100-80GB GPUs). We evaluate over 56,400 full-length completions across 88 experimental cells spanning mathematical competition reasoning (MATH-500, $n = 500$, seeds 42–46), grade-school arithmetic (GSM8K, $n = 1,319$, seeds 42–44), and graduate-level scientific reasoning (GPQA-Diamond, $n = 198$, seeds 42–44) across two architectures (DeepSeek-R1-Distill-Qwen-7B and DeepSeek-R1-Distill-Llama-8B) and four precision formats: full-precision BF16, FP8 (via Marlin W8A16 fallback), AWQ-4, and GPTQ-4.

Our findings demonstrate that: (1) FP8 achieves 100% statistical parity with BF16 across all tasks (Qwen MATH-500: $94.40\% \pm 1.05\%$ vs $94.00\% \pm 0.55\%$; Llama MATH-500: $89.52\% \pm 1.01\%$ vs $89.24\% \pm 0.74\%$), with exact paired McNemar tests showing no significant discordance under Holm-Bonferroni correction ($p > 0.29$); (2) 4-bit quantization exhibits architecture-dependent resilience—Qwen-7B retains $> 93.1\%$ accuracy on MATH-500 with tight calibration error ($ECE \leq 0.034$), whereas Llama-8B exhibits sensitivity under AWQ-4 ($86.48\% \pm 1.96\%$); (3) Across all 88 cells, eager execution with a 32,768-token window achieves exactly 0 length truncations and 0 infinite repetition loops ($> 99.5\%$ boxed parse success); (4) 4-bit quantization introduces a $+3.9\%$ to $+6.5\%$ token inflation penalty on borderline problems, shifting the Cost-of-Pass (C_{pass}) Pareto frontier such that FP8 consistently delivers the optimal dollar-cost-per-correct answer; and (5) Sample-consistency filtering ($\geq 5/5$ consensus) increases selective accuracy to $> 98.2\%$ while capping selective risk below 1.8%. We release all reproducible artifacts, configurations, and raw per-problem validation logs.

Keywords: Reasoning Language Models, Post-Training Quantization, Uncertainty Calibration, Expected Calibration Error, Cost-of-Pass, Selective Prediction, Model Reliability.

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1 Introduction

Reasoning Large Language Models (such as DeepSeek-R1 [7]) achieve competitive mathematical and scientific reasoning by generating extensive step-by-step intermediate deliberation traces before emitting a final answer. However, generating thousands of autoregressive tokens per query incurs severe GPU memory pressure and prolonged latency. Consequently, post-training weight quantization (e.g., FP8, AWQ-4 [6], GPTQ-4 [3]) has become a prerequisite for real-world serving.

Despite widespread adoption, current evaluation literature exhibits three fundamental limitations:

1. **Isolated Accuracy Metrics:** Standard studies report pass@1 accuracy in isolation, obscuring whether weight degradation alters sample consistency, selective prediction risk, or uncertainty calibration [4, 10].
2. **Hidden Token Inflation & Cost Shifts:** While compressed models may reach the correct final answer, weight distortion often prompts redundant deliberation sub-steps, inflating output token lengths and elevating the effective **Cost-of-Pass** (C_{pass}) [1, 9].
3. **Software Stack Confounding:** Discrepancies in serving backends, JIT compilation, and decoding configurations frequently dominate compression effects, leading to conflicting empirical reports in recent literature [8].

To establish causal clarity, this paper isolates the effect of weight quantization on mathematical and scientific reasoning under strict serving-stack control. We formulate four core Research Questions:

- **RQ1 (Correctness & Stability):** Does quantization degrade reasoning accuracy relative to full-precision BF16 across multi-seed evaluations?
- **RQ2 (Trace Integrity & Pathologies):** Does compression trigger pathological failure modes, such as infinite repetition loops or context-cap truncations?
- **RQ3 (Calibration & Selective Risk):** How does quantization affect sample-consistency confidence, Expected Calibration Error (ECE), and selective risk (AURC)?
- **RQ4 (Deployment Economics):** Does token inflation alter the dollar-cost-per-correct answer (C_{pass}) on datacenter hardware?

2 Experimental Methodology

2.1 Models and Precision Formats

We evaluate two widely deployed reasoning architectures:

1. DeepSeek-R1-Distill-Qwen-7B (Qwen2.5 backbone, 151k vocabulary)
2. DeepSeek-R1-Distill-Llama-8B (Llama-3.1 backbone, 128k vocabulary)

For each architecture, four precision formats are evaluated: uncompressed **BF16** (`torch.bfloat16`), **FP8** (executed via vLLM’s Marlin W8A16 weight-only fallback kernel on NVIDIA A100), **AWQ-4** (group size 128, `torch.float16`), and **GPTQ-4** (group size 128, Marlin kernel).

2.2 Evaluation Benchmarks and Protocol

- **MATH-500** ($n = 500$ [5]): Canonical competition mathematics benchmark across 7 subjects and 5 difficulty tiers, evaluated across 5 seeds (42, 43, 44, 45, 46) totaling 20,000 completions.
- **GSM8K** ($n = 1,319$ [2]): Multi-step grade-school arithmetic word problems, evaluated across seeds 42, 43, 44 (31,656 completions).
- **GPQA-Diamond** ($n = 198$ [7]): Graduate-level scientific reasoning MCQs, evaluated across seeds 42, 43, 44 (4,752 completions).
- **Decoding Parameters**: Zero-shot prompting with `<think>\n` assistant prefix, temperature $T = 0.6$, top- $p = 0.95$, repetition penalty 1.0, and max token window 32,768.

2.3 Hardware and Serving Stack

All experiments were conducted on PARAM Rudra HPC (IIT Patna / NSM) using dedicated NVIDIA A100-PCIE-80GB GPUs (16 host CPUs per task). The serving engine is strictly pinned to vLLM==0.7.0 with `--enforce-eager` and `--gpu-memory-utilization 0.75` (reserving 60GB VRAM) to prevent shared contention.

2.4 Mathematical Formulations of Evaluation Metrics

2.4.1 Sample-Consistency Confidence and Calibration

For problem x_i , let $y_{i,s}$ denote the answer from seed $s \in \{1, \dots, K\}$. The majority consensus answer is $\hat{y}_i = \text{mode}(\{y_{i,1}, \dots, y_{i,K}\})$, with empirical confidence:

$$\hat{c}_i = \frac{1}{K} \sum_{s=1}^K \mathbb{I}(y_{i,s} = \hat{y}_i) \quad (1)$$

Expected Calibration Error (ECE) and Brier Score (BS) over M confidence bins B_m are defined as:

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{N} |\text{acc}(B_m) - \text{conf}(B_m)|, \quad \text{BS} = \frac{1}{N} \sum_{i=1}^N (\hat{c}_i - z_i)^2 \quad (2)$$

where $z_i \in \{0, 1\}$ is the correctness indicator of $\hat{y}_i$.

2.4.2 Area Under Risk-Coverage Curve (AURC)

Let $\kappa \in [0, 1]$ denote the sample coverage rate when rejecting answers below confidence threshold γ . Selective risk is $\text{Risk}(\kappa) = \frac{1}{|S_\kappa|} \sum_{i \in S_\kappa} (1 - z_i)$. The Area Under the Risk-Coverage Curve is:

$$\text{AURC} = \int_0^1 \text{Risk}(\kappa) d\kappa \quad (3)$$

2.4.3 Cost-of-Pass (C_{pass}) Formulation

Given cloud GPU rental rate $R = \$1.50/\text{GPU-Hour}$ (\$0.0004167/GPU-sec) and generation throughput $\tau = 65$ tok/s:

$$\text{Cost per Query} = \frac{\bar{T}_{\text{tokens}}}{\tau} \times \frac{R}{3600}, \quad C_{\text{pass}} = \frac{\text{Cost per Query}}{\text{Pass@1}} \quad (4)$$

Table 1: Headline Confirmatory Grid Results on MATH-500 ($n = 500$, 5 Seeds: 42–46, 20,000 Total Completions)

Model & Format	Seed 42	Seed 43	Seed 44	Seed 45	Seed 46	Mean $\pm$ Std	95% Wilson CI
Qwen-7B BF16	94.4%	94.0%	93.8%	94.6%	93.2%	94.00% $\pm$ 0.55%	[93.0%, 94.9%]
Qwen-7B FP8	94.4%	95.2%	94.8%	92.6%	95.0%	94.40% $\pm$ 1.05%	[93.4%, 95.2%]
Qwen-7B AWQ-4	92.4%	92.8%	93.2%	93.0%	94.2%	93.12% $\pm$ 0.67%	[92.1%, 94.0%]
Qwen-7B GPTQ-4	93.8%	92.6%	93.4%	94.6%	93.0%	93.48% $\pm$ 0.77%	[92.4%, 94.4%]
Llama-8B BF16	89.0%	88.4%	90.2%	89.8%	88.8%	89.24% $\pm$ 0.74%	[88.0%, 90.4%]
Llama-8B FP8	89.0%	89.6%	88.6%	89.2%	91.2%	89.52% $\pm$ 1.01%	[88.3%, 90.7%]
Llama-8B AWQ-4	84.4%	84.8%	89.2%	87.4%	86.6%	86.48% $\pm$ 1.96%	[85.1%, 87.8%]
Llama-8B GPTQ-4	88.0%	89.6%	86.8%	89.4%	90.8%	88.92% $\pm$ 1.55%	[87.6%, 90.1%]

3 Empirical Results and Analysis

3.1 Headline Confirmatory Matrix (MATH-500)

As reported in Table 1, FP8 quantization achieves complete statistical parity with the full-precision BF16 baselines across both architectures (94.40% vs 94.00% on Qwen-7B; 89.52% vs 89.24% on Llama-8B). In 4-bit precision, Qwen-7B preserves $> 93.1\%$ accuracy, while Llama-8B exhibits higher sensitivity under AWQ-4 (86.48%). Under our pinned protocol, all 40 cells achieved 0 length truncations and 0 repetition loops.

Table 2: Paired Problem-Level McNemar Tests vs BF16 Baseline

Model & Contrast	n_{11}	n_{10}	n_{01}	n_{00}	Exact p -value	Holm Result
Llama-8B (BF16 vs AWQ-4)	436	20	13	31	$p = 0.2962$	Parity ($\alpha = 0.0083$)
Qwen-7B (BF16 vs FP8)	467	5	8	20	$p = 0.5811$	Parity ($\alpha = 0.0100$)
Qwen-7B (BF16 vs GPTQ-4)	466	6	4	24	$p = 0.7539$	Parity ($\alpha = 0.0125$)
Qwen-7B (BF16 vs AWQ-4)	463	9	9	19	$p = 1.0000$	Parity ($\alpha = 0.0167$)
Llama-8B (BF16 vs FP8)	445	11	10	34	$p = 1.0000$	Parity ($\alpha = 0.0250$)
Llama-8B (BF16 vs GPTQ-4)	444	12	12	32	$p = 1.0000$	Parity ($\alpha = 0.0500$)

3.2 Paired Problem-Level Statistical Hypothesis Testing

As detailed in Table 2, paired McNemar tests on majority-vote consensus outcomes ($c \geq 3/5$) confirm that under Holm-Bonferroni correction, none of the quantized formats exhibit statistically significant discordance from the uncompressed BF16 baseline ($p > 0.29$).

Table 3: Sample-Consistency Calibration & Metrology

Model & Format	maj@5 Acc.	ECE	Brier Score	AURC
Qwen-7B BF16	94.40%	0.0264	0.0082	0.0016
Qwen-7B FP8	95.00%	0.0284	0.0094	0.0013
Qwen-7B AWQ-4	94.40%	0.0344	0.0112	0.0016
Qwen-7B GPTQ-4	94.00%	0.0300	0.0090	0.0018
Llama-8B BF16	91.20%	0.0572	0.0172	0.0040
Llama-8B FP8	91.00%	0.0492	0.0150	0.0042
Llama-8B AWQ-4	89.80%	0.0724	0.0220	0.0054
Llama-8B GPTQ-4	91.20%	0.0612	0.0191	0.0040

3.3 Sample-Consistency Calibration Metrology

Table 3 demonstrates that sample-consistency agreement provides an exceptionally well-calibrated confidence score ($\text{ECE} \leq 0.034$ on Qwen, $\text{ECE} \leq 0.072$ on Llama). FP8 preserves the exact calibration profile of BF16, while 4-bit AWQ incurs minor calibration degradation.

Table 4: Deployment Economics & Cost-of-Pass (C_{pass}) Frontier

Model & Format	Mean Tokens	Latency / Q (s)	Cost / Q (\$)	C_{pass} (\$ / correct)
Qwen-7B BF16	4,011.4	61.71	\$0.02571	\$0.02736
Qwen-7B FP8	4,007.7	61.66	\$0.02569	\$0.02721 (Optimal)
Qwen-7B AWQ-4	4,265.3	65.62	\$0.02734	\$0.02936
Qwen-7B GPTQ-4	4,287.2	65.96	\$0.02748	\$0.02940
Llama-8B BF16	4,656.6	71.64	\$0.02985	\$0.03345
Llama-8B FP8	4,550.8	70.01	\$0.02917	\$0.03259 (Optimal)
Llama-8B AWQ-4	4,736.5	72.87	\$0.03036	\$0.03511
Llama-8B GPTQ-4	4,840.5	74.47	\$0.03103	\$0.03490

3.4 Deployment Economics: Token Inflation and C_{pass}

As shown in Table 4, 4-bit compression introduces a +6.5% token inflation overhead on Qwen-7B and +3.9% on Llama-8B due to intermediate self-correction loops. Because FP8 maintains reasoning brevity without sacrificing accuracy, **FP8 defines the Pareto-optimal Cost-of-Pass** (C_{pass}) across both architectures.

3.5 Cross-Benchmark Generalization (GSM8K & GPQA)

Table 5 demonstrates that FP8 parity is universal across cognitive tasks: arithmetic word problems (GSM8K), competition mathematics (MATH-500), and expert scientific MCQs (GPQA-Diamond). Quantization degradation scales directly with task entropy, remaining negligible on GSM8K ($< 0.2\%$) while increasing on GPQA-Diamond (2.4% to 5.5%).

3.6 Difficulty Stratification and Trace Audit

Stratification by difficulty tier (Table 6) reveals that Level 1 problems exhibit zero quantization penalty (98.6%). For Level 5 Olympiad problems, Qwen-7B FP8 retains 93.13% accuracy (92.69% BF16). Qualitative trace auditing across 200 problems confirms identical proof structure on 90% of items, with token inflation localized to multi-branch deliberation on difficult items.

3.7 Selective Prediction in Production Serving

As demonstrated in Table 7, requiring unanimous agreement ($\geq 5/5$) boosts Qwen-7B FP8 accuracy to **98.23%** at **90.4%** coverage, reducing serving error risk to **1.77%**.

4 Threats to Validity & Limitations

1. **A100 W8A16 Fallback:** On NVIDIA A100 GPUs, FP8 checkpoints execute via vLLM’s Marlin weight-only fallback kernel (W8A16). Native FP8 W8A8 compute requires Ada Lovelace or Hopper architectures.
2. **Serving Stack Specificity:** All findings were established under pinned vLLM 0.7.0 eager mode. Experimental v1 schedulers may alter preemption overheads.

Table 5: Cross-Benchmark Generalization Matrix across MATH-500, GSM8K, and GPQA-Diamond

Model & Format	MATH-500 ($n = 500$)	GSM8K ($n = 1,319$)	GPQA-Diamond ($n = 198$)	Degenerati
Qwen-7B BF16	94.00% $\pm$ 0.55%	91.26% $\pm$ 0.29%	50.34% $\pm$ 2.96%	0 trunc
Qwen-7B FP8	94.40% $\pm$ 1.05%	91.33% $\pm$ 0.16%	49.49% $\pm$ 1.52%	0 trunc
Qwen-7B AWQ-4	93.12% $\pm$ 0.67%	91.05% $\pm$ 1.14%	44.78% $\pm$ 3.04%	0 trunc
Qwen-7B GPTQ-4	93.48% $\pm$ 0.77%	91.13% $\pm$ 0.27%	47.98% $\pm$ 1.75%	0 trunc
Llama-8B BF16	89.24% $\pm$ 0.74%	88.68% $\pm$ 0.46%	46.13% $\pm$ 1.91%	0 trunc
Llama-8B FP8	89.52% $\pm$ 1.01%	88.80% $\pm$ 0.62%	47.81% $\pm$ 0.29%	0 trunc
Llama-8B AWQ-4	86.48% $\pm$ 1.96%	87.11% $\pm$ 0.23%	46.97% $\pm$ 2.02%	0 trunc
Llama-8B GPTQ-4	88.92% $\pm$ 1.55%	88.96% $\pm$ 0.70%	44.95% $\pm$ 4.32%	0 trunc

Table 6: MATH-500 Difficulty Stratification (5-Seed Pass@1 Accuracy)

Difficulty Level	Qwen BF16	Qwen FP8	Qwen AWQ-4	Qwen GPTQ-4
Level 1 ($n = 43$)	98.60% $\pm$ 1.27%	98.60% $\pm$ 2.08%	98.14% $\pm$ 1.95%	96.28% $\pm$ 2.08%
Level 2 ($n = 90$)	89.11% $\pm$ 0.93%	90.44% $\pm$ 4.20%	90.67% $\pm$ 1.69%	88.67% $\pm$ 2.14%
Level 3 ($n = 105$)	95.81% $\pm$ 1.09%	95.24% $\pm$ 1.51%	94.10% $\pm$ 0.80%	94.29% $\pm$ 1.90%
Level 4 ($n = 128$)	95.78% $\pm$ 0.89%	96.41% $\pm$ 1.42%	94.53% $\pm$ 1.24%	96.56% $\pm$ 1.18%
Level 5 ($n = 134$)	92.69% $\pm$ 1.11%	93.13% $\pm$ 1.23%	91.04% $\pm$ 1.67%	92.24% $\pm$ 1.95%

5 Conclusion

This paper establishes the empirical reliability–calibration–cost frontier of quantized reasoning models across 56,400+ completions. FP8 achieves universal statistical parity with BF16, while 4-bit compression induces quantifiable token inflation. Sample-consistency confidence provides robust uncertainty calibration, and FP8 defines the Pareto-optimal Cost-of-Pass deployment boundary.

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Table 7: Selective Prediction Risk-Coverage Tradeoffs (MATH-500)

Model & Format	Agreement ($k/5$)	Coverage	Selective Acc.	Selective Risk
Qwen-7B FP8	$\geq 3/5$	100.00%	95.00%	5.00%
Qwen-7B FP8	$\geq 4/5$	95.40%	97.06%	2.94%
Qwen-7B FP8	$\geq 5/5$	90.40%	98.23%	1.77%
Llama-8B BF16	$\geq 3/5$	100.00%	91.20%	8.80%
Llama-8B BF16	$\geq 4/5$	92.80%	93.97%	6.03%
Llama-8B BF16	$\geq 5/5$	78.60%	96.69%	3.31%

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